

Basic Theory of Semiconductors and P-N Junctions

- ☐ **PN Junction**

- ☐ **MS Junction**

PN Junction

□ PN Junction Electrostatics

Built-in Potential, Depletion Approximation

□ Current-Voltage Characteristics

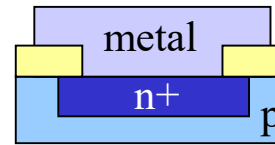
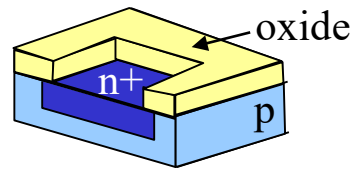
Ideal I - V Characteristics

□ Deviation from the Ideal

Breakdown, High-level Injection

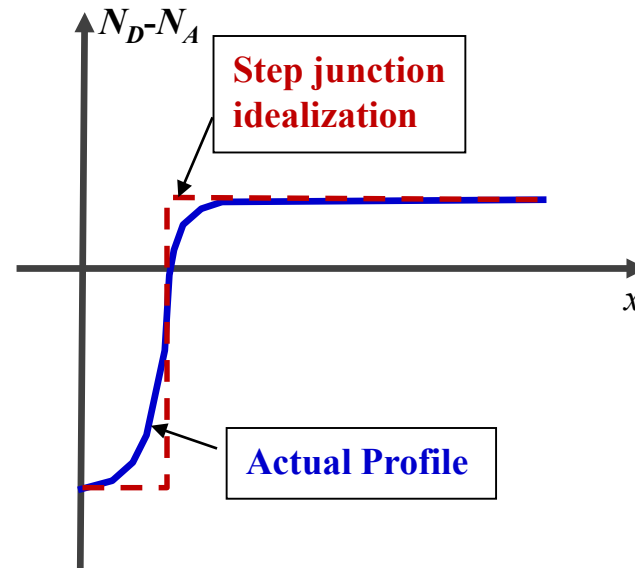
PN Junction Basics

□ Physical View of a PN junction



Cross-section

□ Step Junction

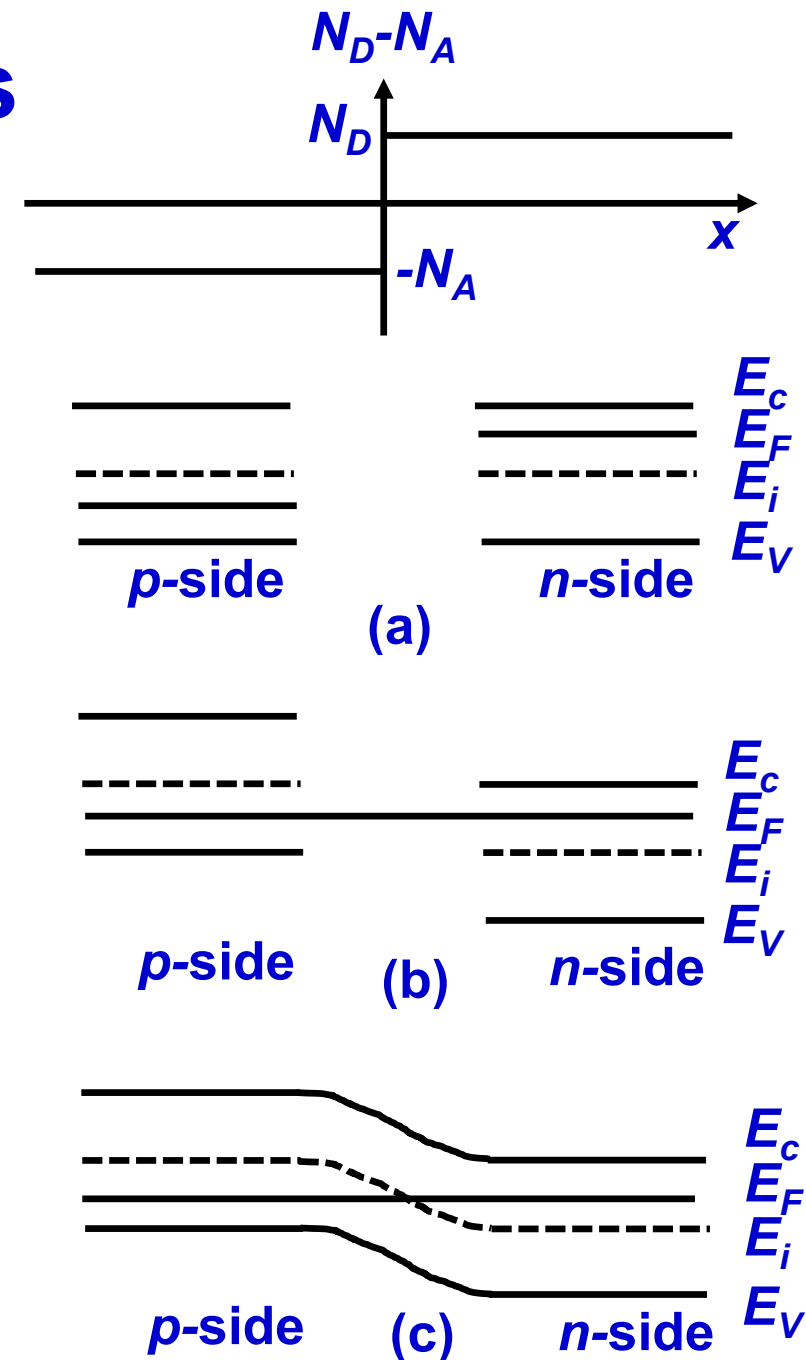


PN Junction Electrostatics

□ Qualitative Analysis:

Thermal Equilibrium

- Energy Band Diagram at Thermal Equilibrium
 - ✓ Fermi level should be aligned
 - ✓ Vacuum level (and hence E_C , E_V , and E_i) should be continuous



PN Junction Electrostatics

□ Qualitative Analysis: Thermal Equilibrium

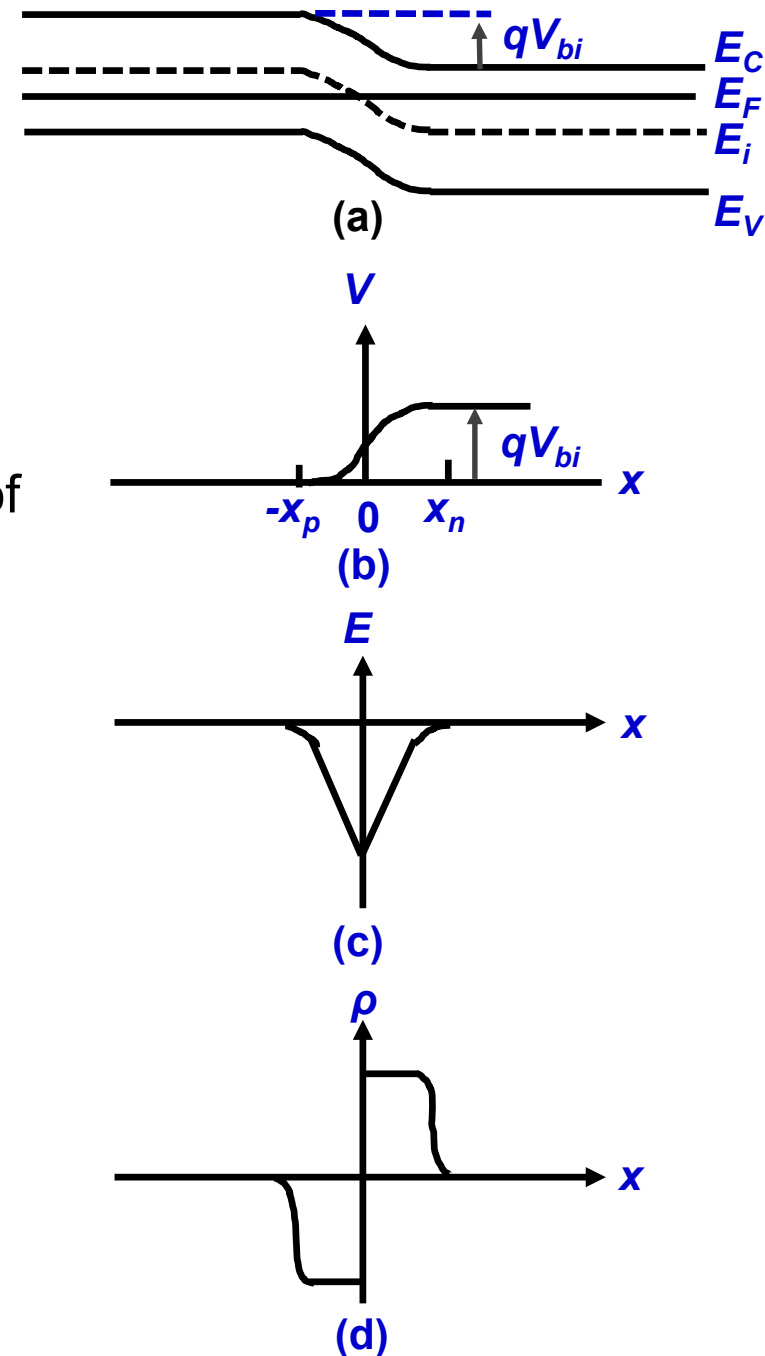
➤ Electrostatic Variables at Thermal Equilibrium

- ✓ The V vs x should have the same functional form of “upside down” of E_C (E_V , or E_i)
- ✓ The E vs x can be drawn from the derivation of V over x ($E = -dV/dx$)
- ✓ The ρ vs x can be drawn from the slope of E

“Built-In Potential”

Since $qV_{bi} = (E_i - E_{Fp}) + (E_{Fn} - E_i)$, we have

$$V_{bi} = \frac{(E_i - E_{Fp}) + (E_{Fn} - E_i)}{q} = \frac{kT}{q} \ln\left(\frac{N_D N_A}{n_i^2}\right)$$

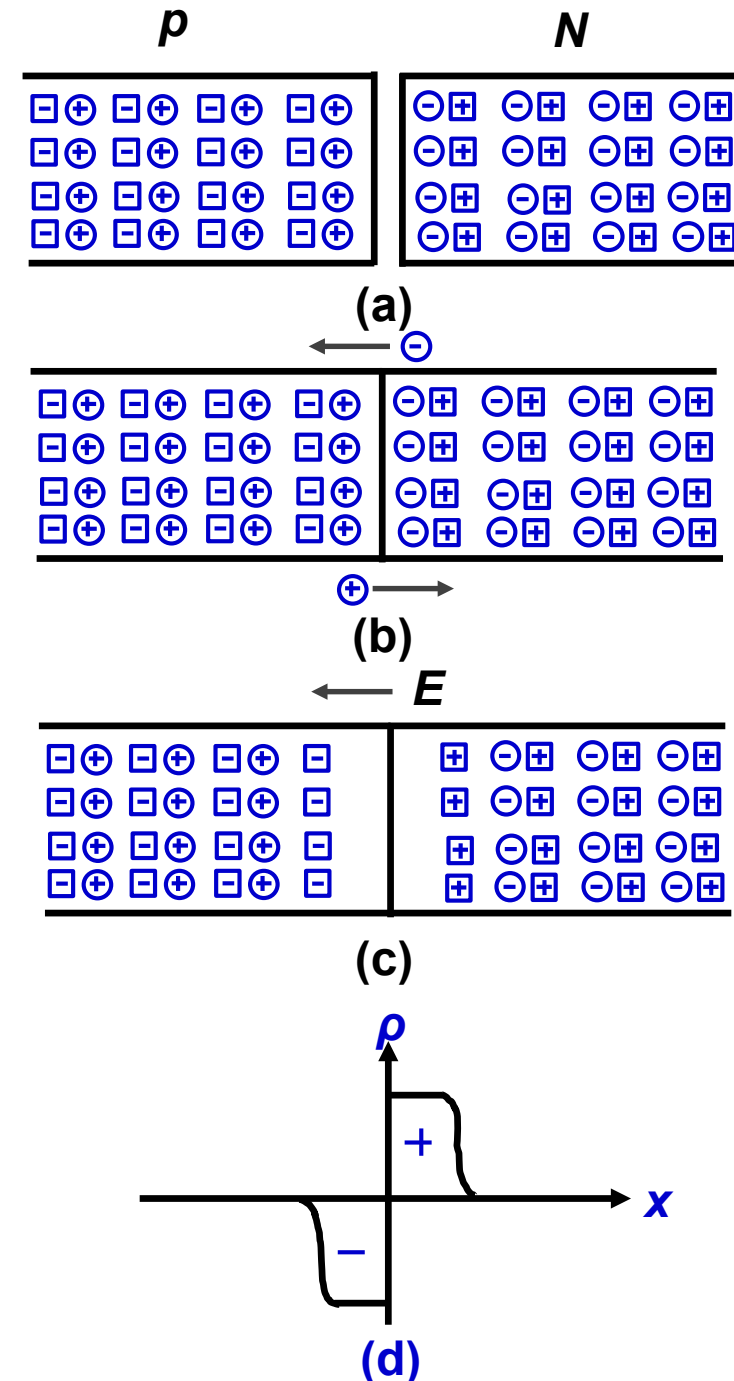


PN Junction Electrostatics

□ Qualitative Analysis: Thermal Equilibrium

➤ Depletion region -- how charge redistributed?

- ✓ Electrons diffusion from n-region to p-region; while holes diffusion from p-region to n-region.
- ✓ Unbalanced dopants are left and form built-in field which directs from n-region to p-region.
- ✓ Equilibrium condition is met when the electrons and holes diffusions are balanced by their drifts due to built-in field.

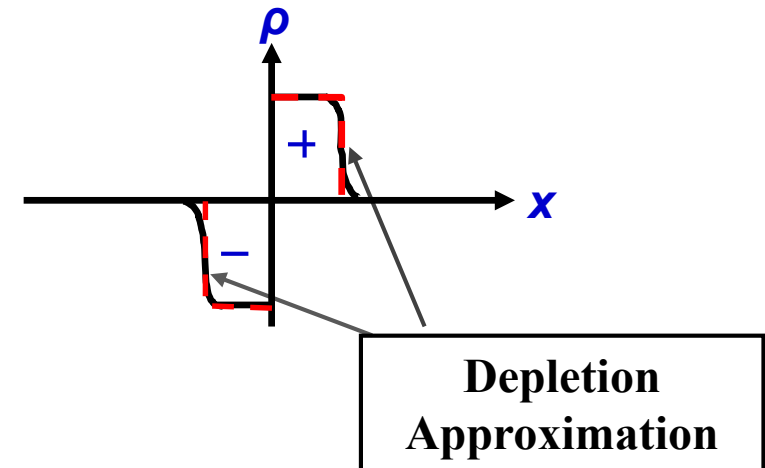
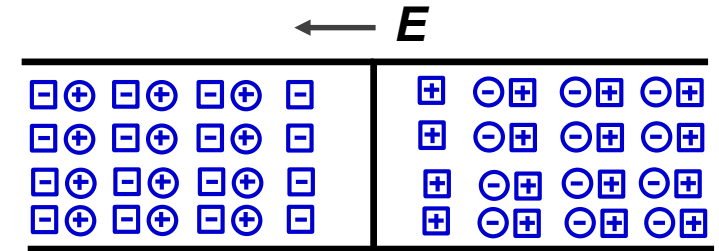


PN Junction Electrostatics

❑ Qualitative Analysis: Thermal Equilibrium

Depletion approximation

- Bulk p and n regions assumed as charge neutral
- Transition region assumed to be depleted of mobile carriers (depletion region)
- Depletion region can be treated as insulator to withstand voltage, but is charged due to unbalanced dopants
- Quite accurate



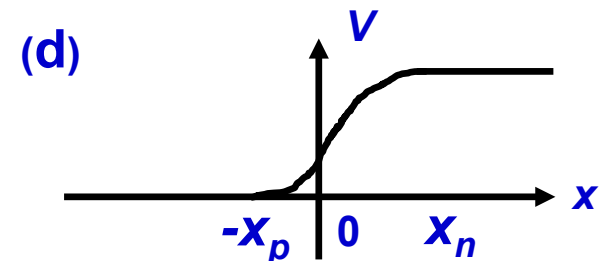
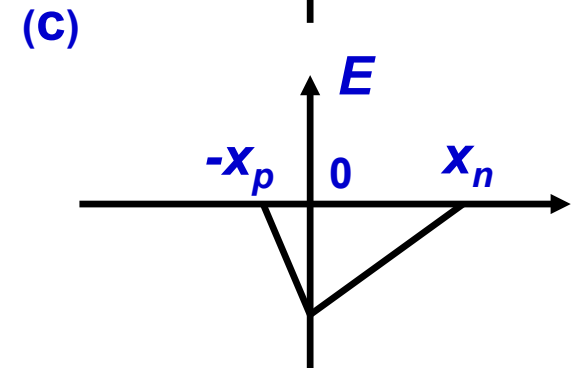
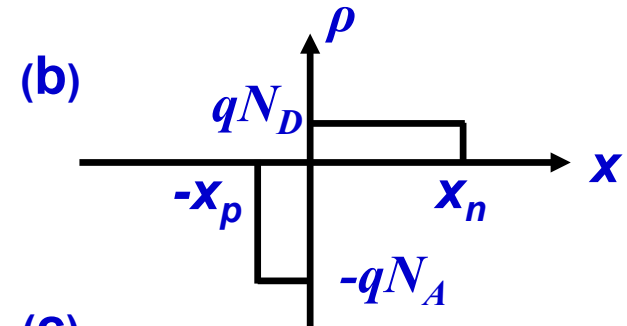
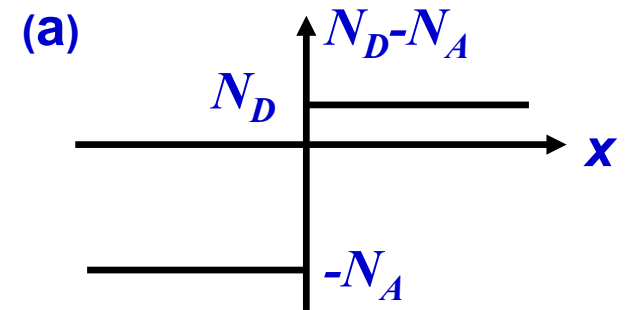
PN Junction Electrostatics

Quantitative Analysis: Thermal Equilibrium

➤ Charge density

- ✓ In n region near the joint, as electrons are depleted (depletion approximation), the region should be positive charged and equal to qN_D
- ✓ Similarly, the p-region near the joint should be negative charged and equal to $-qN_A$
- ✓ So, we have charge density:

$$\rho = \begin{cases} 0 & (x \leq -x_p) \\ -qN_A & (-x_p < x \leq 0) \\ qN_D & (0 < x \leq x_n) \\ 0 & (x > x_n) \end{cases}$$



PN Junction Electrostatics

Quantitative Analysis: Thermal Equilibrium

➤ Electric Field $\frac{dE}{dx} = \frac{\rho}{\epsilon_s \epsilon_0}$

➤ By integration, we have

$$E(x) = -\frac{qN_A}{\epsilon_s \epsilon_0} (x_p + x) \quad \text{for } -x_p \leq x \leq 0$$

$$E(x) = -\frac{qN_D}{\epsilon_s \epsilon_0} (x_n - x) \quad \text{for } 0 \leq x \leq x_n$$

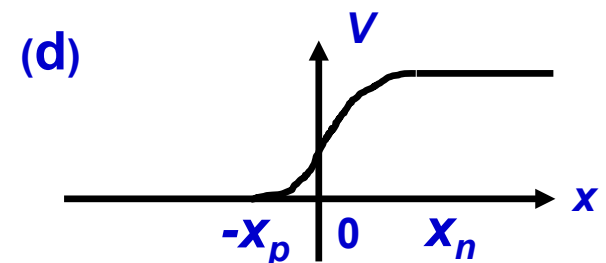
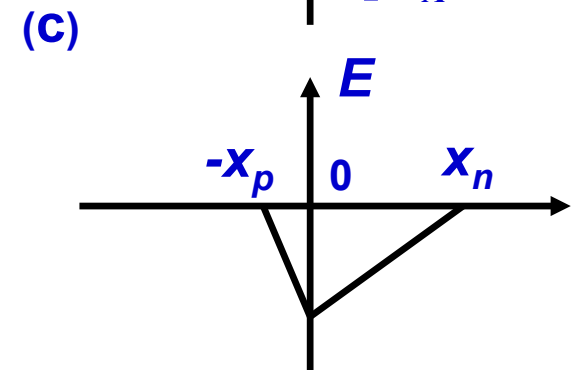
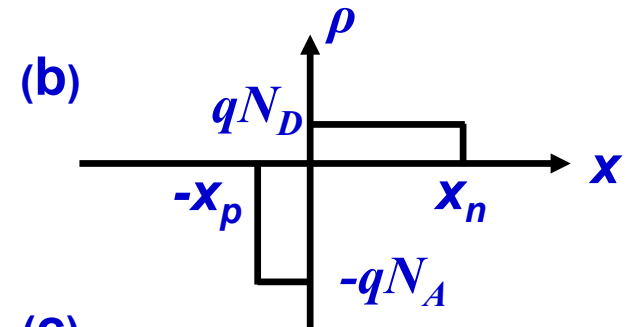
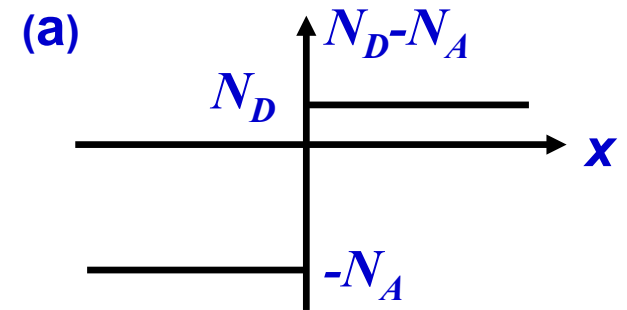
$$E = 0 \quad \text{for } x > x_n \text{ or } x \leq -x_p$$

➤ A note: at $x=0$, the E-fields equal from above two equations, thus

$$N_A x_p = N_D x_n$$

(can also be obtained from charge neutrality)

$$qN_A x_p = qN_D x_n$$



PN Junction Electrostatics

Quantitative Analysis: Thermal Equilibrium

Electric Potential $E = -\frac{dV}{dx}$

With boundary conditions $V = \begin{cases} 0, & x = -x_p \\ V_{bi}, & x = x_n \end{cases}$

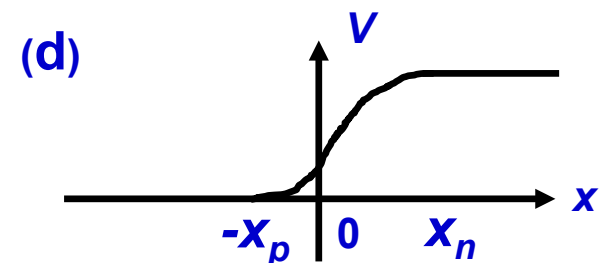
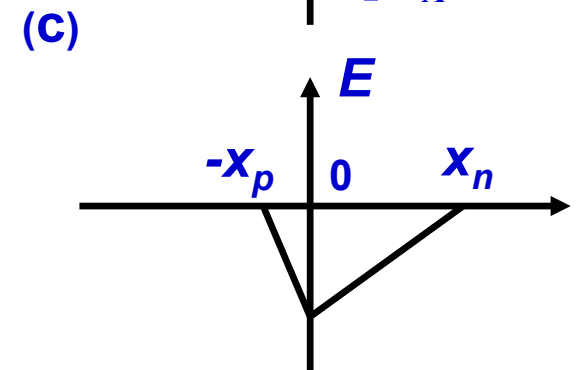
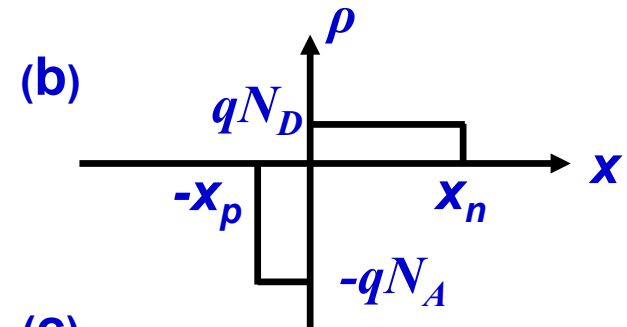
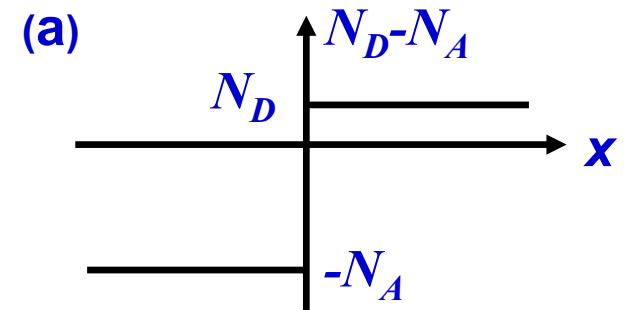
By integration, we have

$$V(x) = \frac{qN_A}{2\epsilon_S\epsilon_0}(x_p + x)^2 \quad \text{for } -x_p \leq x \leq 0$$

$$V(x) = V_{bi} - \frac{qN_D}{2\epsilon_S\epsilon_0}(x_n - x)^2 \quad \text{for } 0 \leq x \leq x_n$$

A note: from electrical potential continuity at $x=0$, one obtains

$$V_{bi} = \frac{qN_A}{2\epsilon_S\epsilon_0}x_p^2 + \frac{qN_D}{2\epsilon_S\epsilon_0}x_n^2$$



PN Junction Electrostatics

Quantitative Analysis: Thermal Equilibrium

Depletion Width

Recall: from continuity of E-field and E-potential at $x=0$, we have

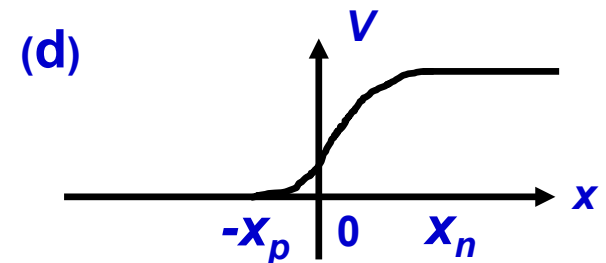
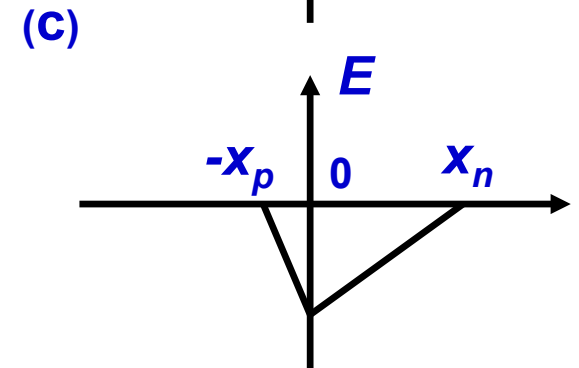
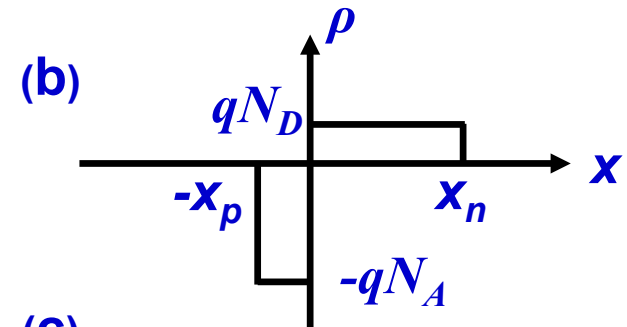
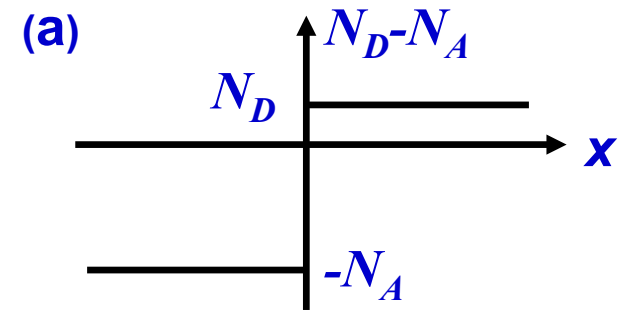
$$N_A x_p = N_D x_n$$

$$V_{bi} = \frac{qN_A}{2\epsilon_S\epsilon_0} x_p^2 + \frac{qN_D}{2\epsilon_S\epsilon_0} x_n^2$$

x_n and x_p can be calculated as below

$$x_n = \left[\frac{2\epsilon_S\epsilon_0}{q} \frac{N_A}{N_D(N_A + N_D)} V_{bi} \right]^{1/2}$$

$$x_p = \left[\frac{2\epsilon_S\epsilon_0}{q} \frac{N_D}{N_A(N_A + N_D)} V_{bi} \right]^{1/2}$$



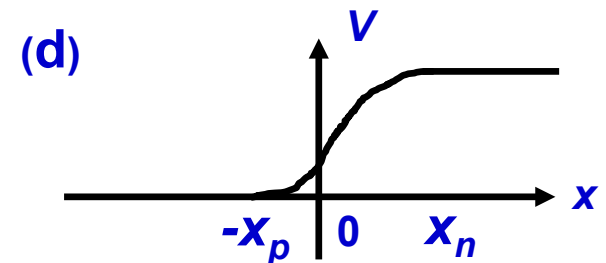
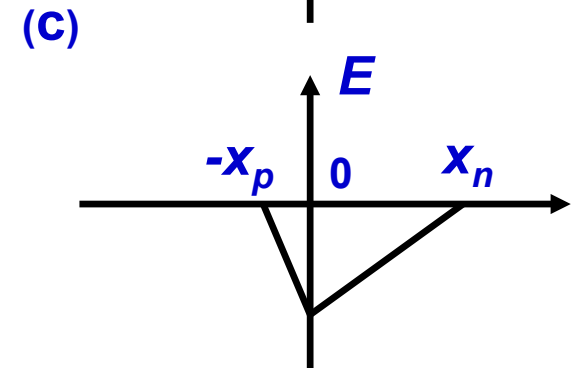
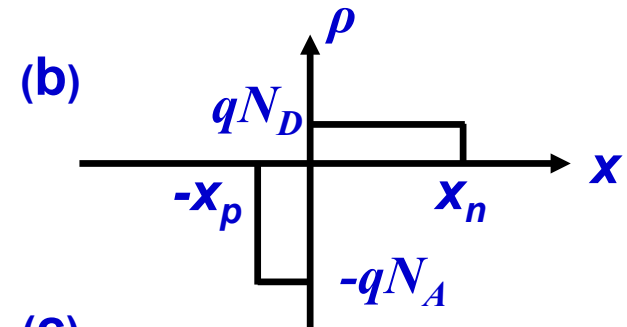
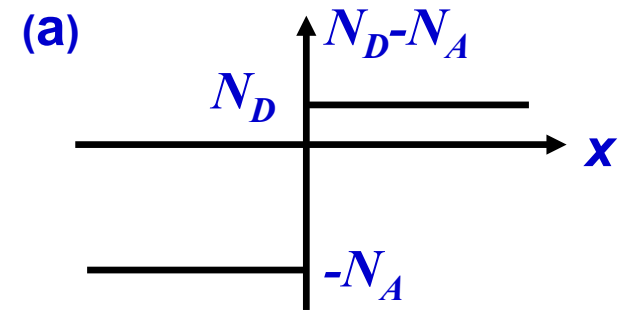
PN Junction Electrostatics

Quantitative Analysis: Thermal Equilibrium

Depletion Width

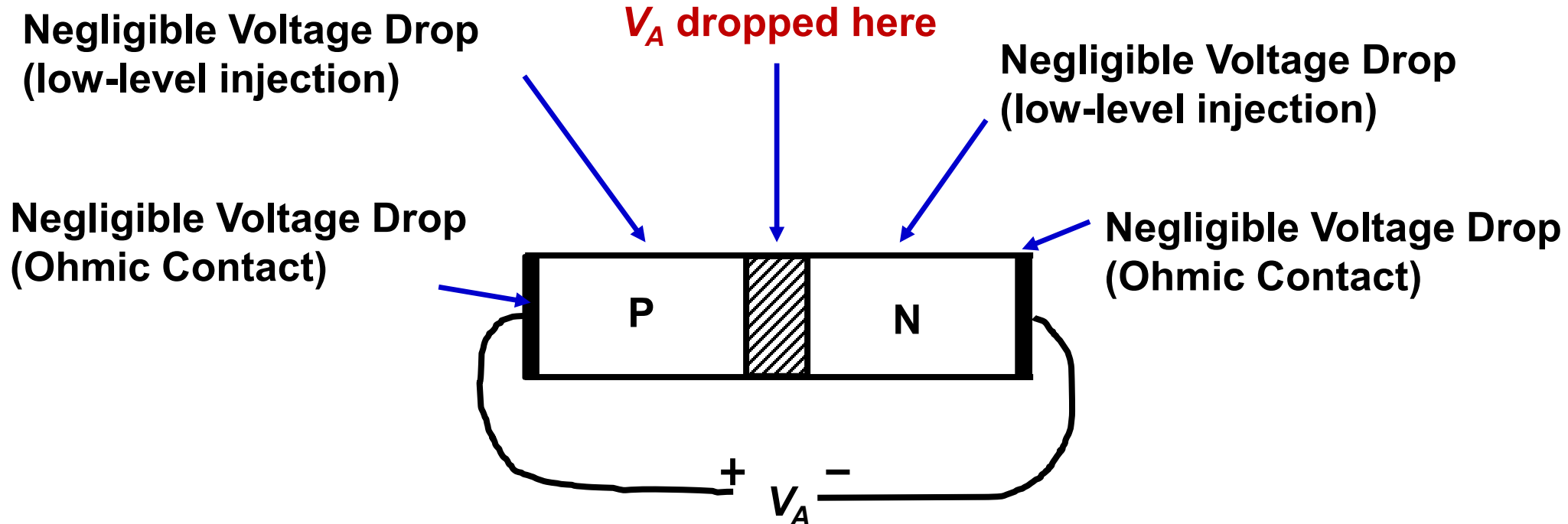
The total depletion width, $W=x_n+x_p$, can be given as below

$$W = x_n + x_p = \left[\frac{2\epsilon_s \epsilon_0}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) V_{bi} \right]^{1/2}$$



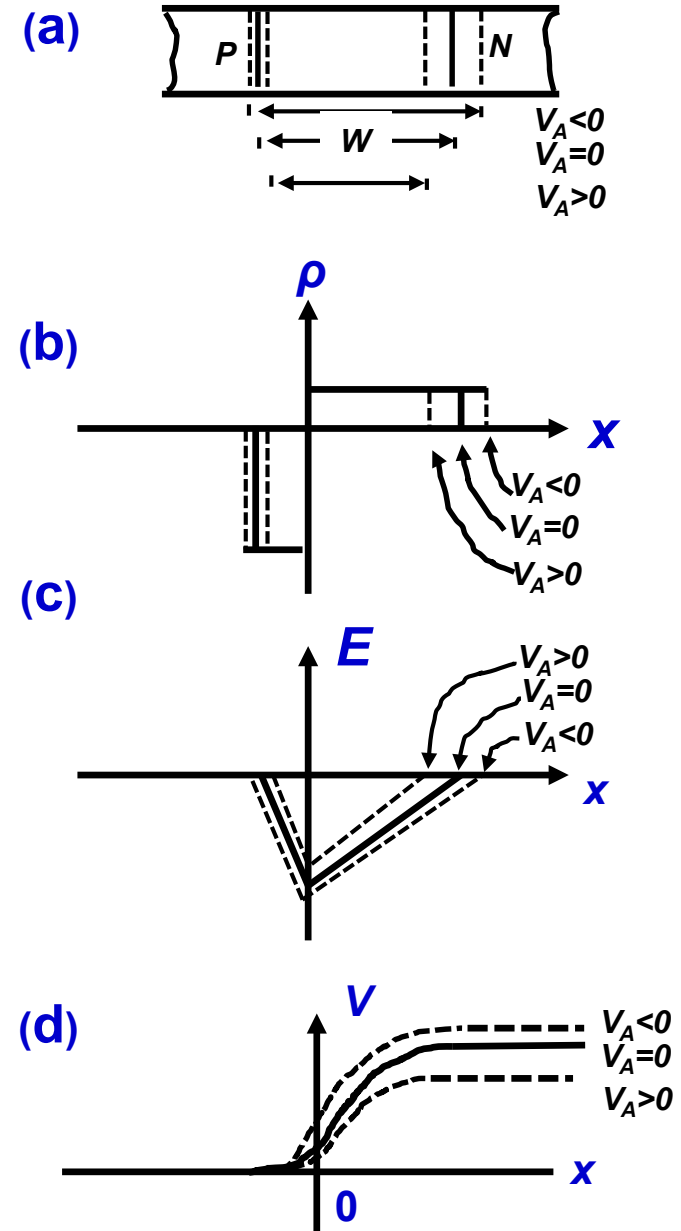
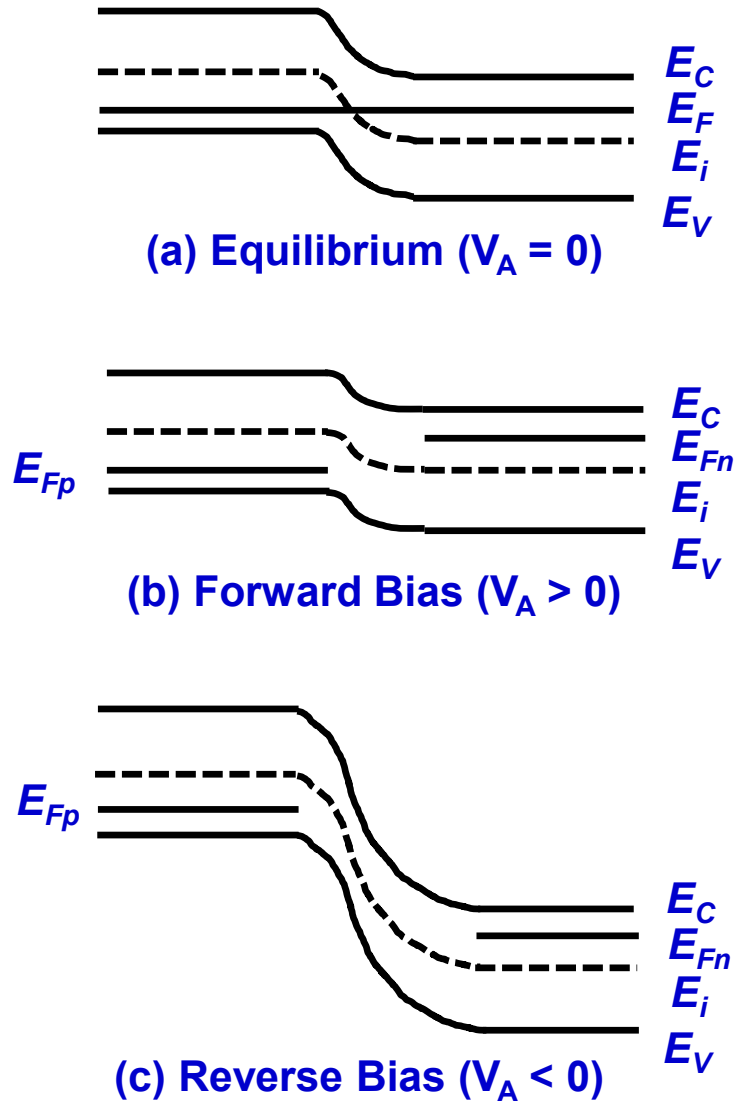
PN Junction Electrostatics

Quantitative Analysis: V_A Applied



PN Junction Electrostatics

Quantitative Analysis: V_A Applied



PN Junction Electrostatics

Quantitative Analysis: V_A Applied

(1) For $x_p \leq x \leq 0$

$$E(x) = -\frac{qN_A}{\epsilon_S \epsilon_0} (x_p + x)$$

$$V(x) = \frac{qN_A}{2\epsilon_S \epsilon_0} (x_p + x)^2$$

$$x_p = \left[\frac{2\epsilon_S \epsilon_0}{q} \frac{N_D}{N_A(N_A + N_D)} (V_{bi} - V_A) \right]^{1/2}$$

(1) For $0 \leq x \leq x_n$

$$E(x) = -\frac{qN_D}{\epsilon_S \epsilon_0} (x_n - x)$$

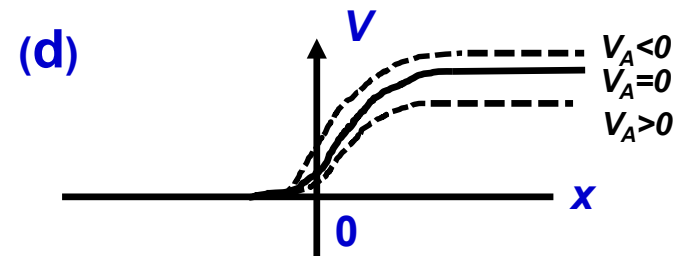
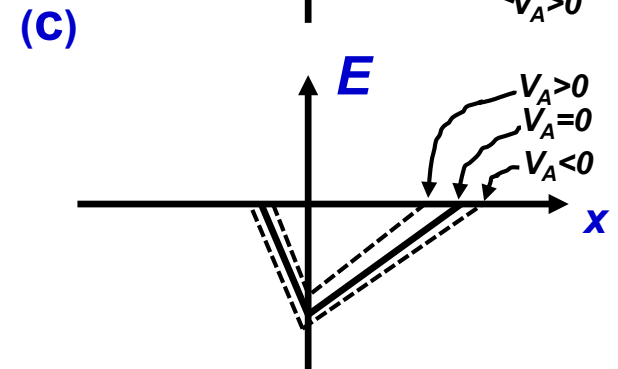
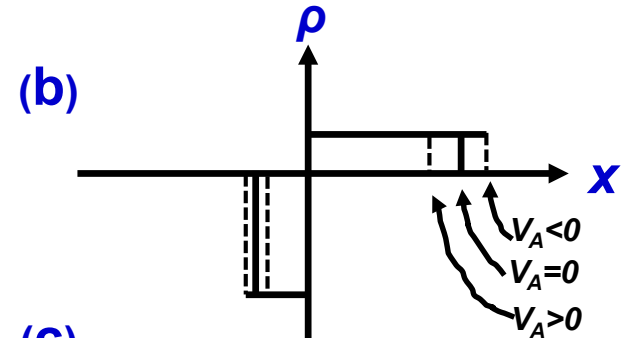
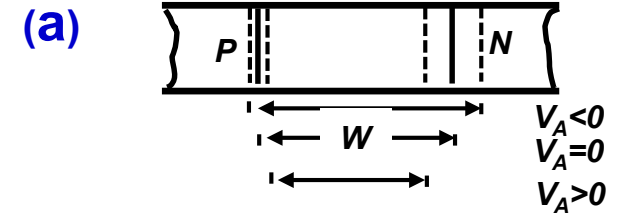
$$V(x) = (V_{bi} - V_A) - \frac{qN_D}{2\epsilon_S \epsilon_0} (x_n - x)^2$$

$$x_n = \left[\frac{2\epsilon_S \epsilon_0}{q} \frac{N_A}{N_D(N_A + N_D)} (V_{bi} - V_A) \right]^{1/2}$$

(3) and the total depletion width W

$$W = x_n + x_p = \left[\frac{2\epsilon_S \epsilon_0}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) (V_{bi} - V_A) \right]^{1/2}$$

Using $(V_{bi} - V_A)$ to replace V_{bi}

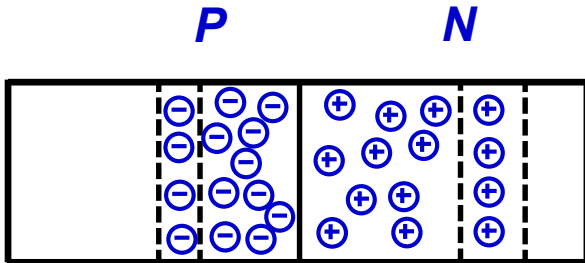
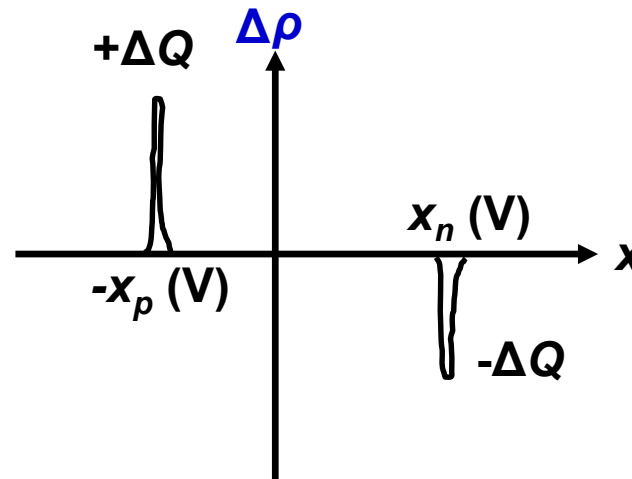
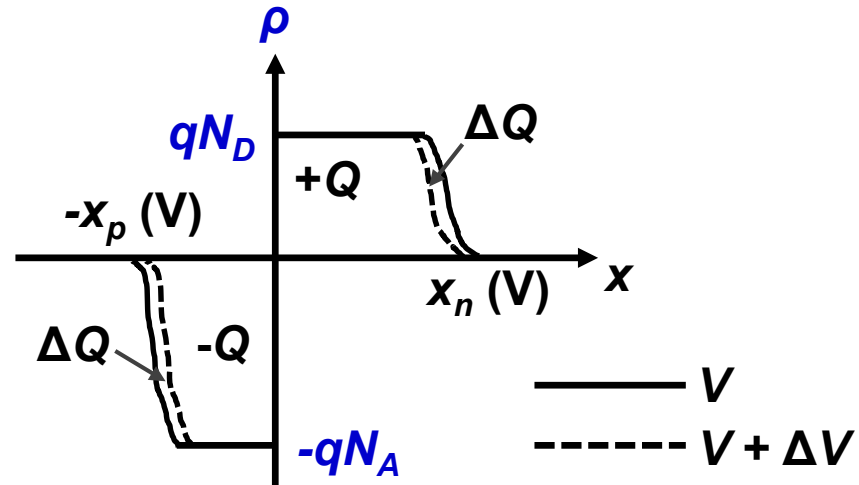


PN Junction Electrostatics

Quantitative Analysis: V_A Applied

Depletion capacitance

$$C = \frac{\epsilon_S \epsilon_0 A}{W}$$



PN Junction Electrostatics

□ Example: Asymmetric Junction

➤ Very often we have N+/P or P+/N junction

➤ For example:

$$n_i/p_i \sim 10^{10}\text{cm}^{-3}$$

$$N_D+/N_A+ \sim 10^{18}\text{cm}^{-3}$$

$$N_D/N_A \sim 10^{15}\text{cm}^{-3}$$

➤ For N+/P, only x_p has to be considered

$$x_p = \left(\frac{2\epsilon_s V_{bi}}{q} \frac{1}{N_A} \right)^{1/2}$$

